

Rigorous Note

Peak-Current Derivations for BJT- and NMOS-Based Self-Oscillating Joule Thief Converters

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1 Introduction and Purpose

This note revisits the peak-current formulas used in the manuscript from a more explicit mathematical standpoint. Starting from a piecewise description of the switching transition that keeps the nearby critical instants distinct, it identifies each approximation step and shows that the manuscript formulas become reductions of a more complete model.

2 Critical Instants Near the Peak

The manuscript compresses several physically distinct instants near the end of the conduction interval into a single marker t_{on} . This section keeps them separate so that the later reductions are explicit.

2.1 Shared Instants

Throughout this note, the generic switch-branch current is denoted i_{switch} : in the BJT converter $i_{\text{switch}} = i_{\text{CE}}$, and in the NMOS converter $i_{\text{switch}} = i_{\text{ds}}$. The following instants are common to both converters:

- t_{on} : switch-current peak, defined by $di_{\text{switch}}/dt|_{t=t_{\text{on}}} = 0$;
- t_1 : primary-current peak, defined by $di_1/dt|_{t=t_1} = 0$;
- t_L : the instant at which $V_{L1}(t_L) = 0$;
- t_{sw} : the instant at which $i_{\text{switch}}(t_{\text{sw}}) = 0$, marking the end of commutation;
- $T_{\text{comm}} \equiv t_{\text{sw}} - t_{\text{on}}$: the commutation interval.

2.2 Device-Specific Peak Markers

For the BJT, t_B denotes the instant at which $dV_{\text{BE}}/dt|_{t=t_B} = 0$ and t_C the instant at which $di_{\text{CE}}/dt|_{t=t_C} = 0$. Since $i_{\text{switch}} = i_{\text{CE}}$ during BJT conduction, and we define

$$t_{\text{on}} \equiv t_C. \quad (1)$$

For the NMOS, t_G denotes the instant at which $dV_{\text{gs}}/dt|_{t=t_G} = 0$ and t_D the instant at which $di_{\text{ds}}/dt|_{t=t_D} = 0$. Analogously,

$$t_{\text{on}} \equiv t_D. \quad (2)$$

The quantities evaluated at the peak instant t_{on} are denoted by a superscript $*$ (for example, $V_{\text{CE}}^* \equiv V_{\text{CE}}(t_{\text{on}})$).

3 Modeling Assumptions and Piecewise Current Function

3.1 Ideal-Coupling Relation

Both derivations use the same coupled-inductor relation. For a pair with turns ratio $1 : a$ and the inverse-coupling convention, the winding voltages are

$$V_{L1} = L_1 \frac{di_1}{dt} - M \frac{di_2}{dt}, \quad (3)$$

$$V_{L2} = L_2 \frac{di_2}{dt} - M \frac{di_1}{dt}. \quad (4)$$

Under ideal coupling, $L_2 = a^2 L_1$ and $M = a L_1$. Substituting into (4):

$$V_{L2} = a^2 L_1 \frac{di_2}{dt} - a L_1 \frac{di_1}{dt} = -a \left(L_1 \frac{di_1}{dt} - a L_1 \frac{di_2}{dt} \right) = -a V_{L1}, \quad (5)$$

which is an exact algebraic identity under ideal coupling, independent of the winding currents. This relation is applied directly in both device sections.

3.2 Exact Three-Segment Current Model

For both converters, KCL at the primary branch gives

$$i_1 \equiv i_{\text{switch}} + i_{\text{D}}. \quad (6)$$

After $t = t_{\text{on}}$, current transfers from the switch branch to the diode branch over a finite interval $[t_{\text{on}}, t_{\text{sw}}]$, during which both branches conduct simultaneously. Let $i_{\text{off}}(t)$ denote the decaying switch current during this interval, satisfying

$$i_{\text{off}}(t_{\text{on}}) = i_{\text{peak}}, \quad i_{\text{off}}(t_{\text{sw}}) = 0. \quad (7)$$

The specific shape of $i_{\text{off}}(t)$ is not assumed, because only these boundary values matter. The exact three-segment model is

$$i_{\text{switch}}(t) = \begin{cases} i_1(t), & 0 \leq t < t_{\text{on}}, \\ i_{\text{off}}(t), & t_{\text{on}} \leq t < t_{\text{sw}}, \\ 0, & t_{\text{sw}} \leq t < T, \end{cases} \quad (8)$$

$$i_{\text{D}}(t) = \begin{cases} 0, & 0 \leq t < t_{\text{on}}, \\ i_1(t) - i_{\text{off}}(t), & t_{\text{on}} \leq t < t_{\text{sw}}, \\ i_1(t), & t_{\text{sw}} \leq t < T. \end{cases} \quad (9)$$

3.3 Reduction to the Two-Segment Model

When the commutation interval is relatively short, $T_{\text{comm}} \ll T_{\text{on}}$, the three-segment model reduces to the two-segment form used in the manuscript:

$$i_{\text{switch}}(t) \approx \begin{cases} i_1(t), & 0 \leq t < t_{\text{on}}, \quad (\text{turn-on phase}) \\ 0, & t_{\text{on}} \leq t < T, \quad (\text{turn-off phase}) \end{cases} \quad (10)$$

$$i_{\text{D}}(t) \approx \begin{cases} 0, & 0 \leq t < t_{\text{on}}, \quad (\text{turn-on phase}) \\ i_1(t), & t_{\text{on}} \leq t < T. \quad (\text{turn-off phase}) \end{cases} \quad (11)$$

The two-segment form is just for convenience to label the two operating states, while the subsequent derivations are not dependent on it. The commutation interval $[t_{\text{on}}, t_{\text{sw}}]$ is neglected, which is a very common simplification in power electronics steady-state analysis and is quantitatively justified in Section 5: $T_{\text{comm}} < 0.2\% T_{\text{on}}$ for the representative operating point.

3.4 Peak Condition in the Reduced Model

In the two-segment model, the derivative di_{switch}/dt maybe ambiguous at the switching corner $t = t_{\text{on}}$. The physically meaningful peak condition is the **left-hand limit** inherited from the turn-on phase. The left-hand limit of di_{switch}/dt therefore equals that of di_1/dt :

$$\left. \frac{di_{\text{switch}}}{dt} \right|_{t=t_{\text{on}}^-} = \lim_{t \rightarrow t_{\text{on}}^-} \frac{di_1}{dt}. \quad (12)$$

In the BJT case, $i_1 = i_{\text{CE}} + i_{\text{D}}$ and $i_{\text{D}} \approx 0$ in the turn-on phase, so (12) becomes $di_{\text{CE}}/dt|_{t=t_{\text{C}}} = 0$. In the NMOS case, the same argument gives $di_{\text{ds}}/dt|_{t=t_{\text{D}}} = 0$. In both cases, the peak formula is obtained by evaluating the device constitutive equations at t_{on} . The behavior after this instant is irrelevant.

4 BJT Peak-Current Derivation

4.1 Governing Equations

Applying KVL to the collector loop and the base loop during BJT conduction gives

$$V_{\text{L1}} = V_{\text{oc}} - V_{\text{CE}}, \quad (13)$$

$$V_{\text{L2}} = V_{\text{oc}} - R_{\text{b}}i_{\text{B}} - V_{\text{BE}}. \quad (14)$$

Eliminating V_{L2} using the ideal-coupling relation (5) gives

$$V_{\text{BE}} + aV_{\text{CE}} = (1 + a)V_{\text{oc}} - R_{\text{b}}i_{\text{B}}. \quad (15)$$

The Ebers-Moll model gives

$$i_{\text{B}} = \frac{I_{\text{s}}}{\beta_{\text{F}}} \left(e^{V_{\text{BE}}/V_{\text{T}}} - 1 \right) + \frac{I_{\text{s}}}{\beta_{\text{R}}} \left(e^{V_{\text{BC}}/V_{\text{T}}} - 1 \right), \quad (16)$$

$$i_{\text{CE}} = I_{\text{s}} \left(e^{V_{\text{BE}}/V_{\text{T}}} - e^{V_{\text{BC}}/V_{\text{T}}} \right) - \frac{I_{\text{s}}}{\beta_{\text{R}}} \left(e^{V_{\text{BC}}/V_{\text{T}}} - 1 \right), \quad (17)$$

where $V_{\text{BC}} = V_{\text{BE}} - V_{\text{CE}}$. Substituting (16) into (15) defines the implicit trajectory constraint

$$F(V_{\text{BE}}, V_{\text{CE}}) \equiv V_{\text{BE}} + aV_{\text{CE}} + R_{\text{b}}i_{\text{B}} - (1 + a)V_{\text{oc}} = 0, \quad (18)$$

which holds throughout the BJT conduction interval.

4.2 Exact Peak Condition

Applying the chain rule to $i_{\text{CE}}(V_{\text{BE}}, V_{\text{CE}})$ and imposing the peak condition $di_{\text{CE}}/dt = 0$ gives

$$\frac{\partial i_{\text{CE}}}{\partial V_{\text{BE}}} \frac{dV_{\text{BE}}}{dt} + \frac{\partial i_{\text{CE}}}{\partial V_{\text{CE}}} \frac{dV_{\text{CE}}}{dt} = 0. \quad (19)$$

The simulated waveform in Fig. 1, presented later in Section 5, shows that in a neighborhood of $t = t_{\text{C}}$, V_{CE} continues to increase while i_{CE} reaches its local maximum. This behavior is also physically consistent with the end of the turn-on interval: as the feedback drive weakens, the collector-current slope collapses and the primary inductor voltage decreases. From the collector-loop relation (13), a decreasing V_{L1} corresponds to an increasing V_{CE} . Hence,

$$\left. \frac{dV_{\text{CE}}}{dt} \right|_{t=t_{\text{C}}} > 0, \quad (20)$$

Therefore V_{BE} can be treated as a local function of V_{CE} near $t = t_{\text{C}}$. Dividing (19) by $dV_{\text{CE}}/dt \neq 0$ gives

$$\left. \frac{dV_{\text{BE}}}{dV_{\text{CE}}} \right|_{t=t_{\text{C}}} = - \frac{\partial i_{\text{CE}}/\partial V_{\text{CE}}}{\partial i_{\text{CE}}/\partial V_{\text{BE}}}. \quad (21)$$

From (17), the partial derivatives evaluated at $t = t_C$ are

$$\frac{\partial i_{CE}}{\partial V_{BE}} = \frac{I_s}{V_T} \left[e^{V_{BE}^*/V_T} - \left(1 + \frac{1}{\beta_R} \right) e^{V_{BC}^*/V_T} \right], \quad (22)$$

$$\frac{\partial i_{CE}}{\partial V_{CE}} = \frac{I_s}{V_T} \left(1 + \frac{1}{\beta_R} \right) e^{V_{BC}^*/V_T}. \quad (23)$$

Substituting into (21) and dividing numerator and denominator by $e^{V_{BE}^*/V_T}$ (noting $V_{BC} - V_{BE} = -V_{CE}$), the exact voltage-plane slope at the peak is

$$\left. \frac{dV_{BE}}{dV_{CE}} \right|_{t=t_C} = - \frac{\left(1 + \frac{1}{\beta_R} \right) e^{-V_{CE}^*/V_T}}{1 - \left(1 + \frac{1}{\beta_R} \right) e^{-V_{CE}^*/V_T}}. \quad (24)$$

4.3 Exact Peak-Voltage Equation

Since $F(V_{BE}, V_{CE}) = 0$ holds throughout the conduction interval, differentiating along the trajectory gives

$$\frac{\partial F}{\partial V_{BE}} \frac{dV_{BE}}{dt} + \frac{\partial F}{\partial V_{CE}} \frac{dV_{CE}}{dt} = 0. \quad (25)$$

Dividing by $dV_{CE}/dt \neq 0$ (established in Section 4.2) at $t = t_C$ gives

$$\left. \frac{\partial F}{\partial V_{CE}} \right|_{t=t_C} = - \left. \frac{dV_{BE}}{dV_{CE}} \right|_{t=t_C} \left. \frac{\partial F}{\partial V_{BE}} \right|_{t=t_C}. \quad (26)$$

From (18) and (16),

$$\left. \frac{\partial F}{\partial V_{BE}} \right|_{t=t_C} = 1 + \frac{R_b I_s}{\beta_F V_T} e^{V_{BE}^*/V_T} + \frac{R_b I_s}{\beta_R V_T} e^{V_{BC}^*/V_T}, \quad (27)$$

$$\left. \frac{\partial F}{\partial V_{CE}} \right|_{t=t_C} = a - \frac{R_b I_s}{\beta_R V_T} e^{V_{BC}^*/V_T}. \quad (28)$$

Substituting (27)–(28) and (24) into (26) yields the exact peak-voltage condition

$$a - \frac{R_b I_s}{\beta_R V_T} e^{V_{BC}^*/V_T} = - \left. \frac{dV_{BE}}{dV_{CE}} \right|_{t=t_C} \left[1 + \frac{R_b I_s}{\beta_F V_T} e^{V_{BE}^*/V_T} + \frac{R_b I_s}{\beta_R V_T} e^{V_{BC}^*/V_T} \right]. \quad (29)$$

4.4 Reduction to the Manuscript Formulas

For typical BJT operating points ($V_{oc} \geq 0.7$ V under an ideal source) in this circuit, V_{CE}^* far exceeds the thermal voltage $V_T \approx 25.85$ mV. Eq. (24) is controlled by

$$\epsilon_s \equiv \left(1 + \frac{1}{\beta_R} \right) e^{-V_{CE}^*/V_T}. \quad (30)$$

Although the load resistance R_L does not appear explicitly in the peak-current derivations, it still affects the overall steady-state performance. At large R_L , the increased voltage swing can require a larger V_{BE} swing. Through magnetic coupling, this can disturb the idealized boundary condition $i_1(0) = 0$. In the typical BJT and NMOS simulations, the analytical model remains accurate for $R_L < 100$ k Ω , whereas substantially larger load resistances lead to larger deviations.

For the parameters used in this note, evaluating the closed-form solution at $V_{oc} = 0.7$ V gives $V_{CE}^* \approx 0.18$ V, so $\epsilon_s < 1.5 \times 10^{-3}$. For larger V_{oc} , the exponential term decreases further. For instance, LTspice gives $V_{CE}^* \approx 0.202$ V at $V_{oc} = 1$ V, so $V_{CE}^*/V_T \approx 7.8$.

Consequently,

$$1 \gg \epsilon_s \approx 0, \quad (31)$$

and Eq. (24) gives

$$\left. \frac{dV_{BE}}{dV_{CE}} \right|_{t=t_C} \approx 0. \quad (32)$$

This is confirmed numerically in Section 5: the exact slope evaluates to -6.13×10^{-4} (Eq. (42)). This approximation sets the right-hand side of Eq. (29) to zero, giving

$$a - \frac{R_b I_s}{\beta_R V_T} e^{V_{BC}^*/V_T} \approx 0, \quad (33)$$

from which

$$V_{BC}^* \approx V_T \ln \left(\frac{a \beta_R V_T}{I_s R_b} \right) \quad (34)$$

and since $V_{BC} = V_{BE} - V_{CE}$, we have

$$V_{BE}^* \approx V_{CE}^* + V_T \ln \left(\frac{a \beta_R V_T}{I_s R_b} \right). \quad (35)$$

This is the voltage relation used in the manuscript. Substituting it into the constraint $F = 0$ yields a transcendental equation in V_{CE}^* alone, which is solved in closed form using the Lambert W function:

$$V_{CE}^* = \xi_1 - V_T W(\xi_2), \quad (36)$$

where

$$\xi_1 = V_{oc} + \frac{I_s R_b}{1+a} \left(\frac{1}{\beta_F} + \frac{1}{\beta_R} \right) - \frac{V_T}{1+a} \left[a + \ln \left(\frac{a V_T \beta_R}{I_s R_b} \right) \right], \quad (37)$$

$$\xi_2 = \frac{a \beta_R}{(1+a) \beta_F} \exp \left(\frac{\xi_1}{V_T} \right). \quad (38)$$

With V_{CE}^* and V_{BE}^* , the peak collector current is obtained by substituting these voltages into the Ebers-Moll model:

$$i_{\text{peak}} \equiv i_{CE}(t_{\text{on}}) \approx I_s \left[e^{V_{BE}^*/V_T} - e^{V_{BC}^*/V_T} \right] - \frac{I_s}{\beta_R} \left[e^{V_{BC}^*/V_T} - 1 \right]. \quad (39)$$

5 LTspice Validation of the BJT Reduction

The LTspice results here are used solely to confirm that the reductions are numerically reasonable for a representative operating point: the nearby critical instants should be close, the commutation interval short relative to T_{on} , and the slope (32) quantitatively small.

5.1 Representative Operating Point

The BJT circuit was simulated with

$$V_{oc} = 1 \text{ V}, \quad R_b = 10 \text{ k}\Omega, \quad L_1 = 1 \text{ mH}, \quad a = 1, \quad k = 0.99,$$

using the transistor model

$$\text{.MODEL myBJT npn}(IS=2.05e-14 \text{ BF}=100 \text{ BR}=1.88).$$

Under this specific `.MODEL` definition, LTspice also uses an Ebers-Moll-based BJT model for simulation.

Figure 1 shows the simulated waveforms zoomed in on the last steady-state pulse, with the key timing markers noted. The key time markers in Table 1 satisfy

$$t_B \approx t_C < t_1 < t_L < t_{\text{sw}}. \quad (40)$$

Table 1: Critical instants extracted from the last steady-state LTspice pulse ($T_{\text{on}} = 11.789 \mu\text{s}$).

Symbol	Definition	Value	Offset from t_C	Offset / T_{on}
t_B	$dV_{\text{BE}}/dt = 0$	$80.791 \mu\text{s}$	0.000 ns	0.000%
$t_C = t_{\text{on}}$	$di_{\text{CE}}/dt = 0$	$80.791 \mu\text{s}$	0.000 ns	0.000%
t_1	$di_1/dt = 0$	$80.804 \mu\text{s}$	13.125 ns	0.111%
t_L	$V_{\text{L1}} = 0$	$80.812 \mu\text{s}$	20.670 ns	0.175%
t_{sw}	$i_{\text{switch}} = 0$	$80.814 \mu\text{s}$	22.867 ns	0.194%

5.2 Quantitative Support for the Approximations

At the collector-current peak, LTspice gives

$$V_{\text{CE}}(t_C) \approx 0.202 \text{ V}, \quad V_{\text{BE}}(t_C) \approx 0.697 \text{ V}. \quad (41)$$

Using these values in (24) gives

$$\left. \frac{dV_{\text{BE}}}{dV_{\text{CE}}} \right|_{t=t_C} \approx -6.13 \times 10^{-4}, \quad (42)$$

The diode current at $t = t_C$ is approximately $5.99 \mu\text{A}$, or 0.057% of $i_{\text{CE}}(t_C) \approx 10.46 \text{ mA}$, confirming that commutation has barely begun at the collector-current peak. At t_1 , the inductor voltages are still

$$V_{\text{L1}}(t_1) \approx 0.766 \text{ V}, \quad V_{\text{L2}}(t_1) \approx -0.743 \text{ V}, \quad (43)$$

so the zero crossing of V_{L1} occurs later, consistent with (40).

Table 2: Approximate analytical predictions versus LTspice extraction.

Quantity	Approximate model	LTspice	Rel. error
V_{CE}^*	0.199 V	$0.202 \text{ V at } t_C$	-1.56%
V_{BE}^*	0.698 V	$0.697 \text{ V at } t_C$	$+0.049\%$
$i_{\text{peak}} = i_{\text{CE}}(t_C)$	10.766 mA	10.460 mA	$+2.92\%$
$i_{\text{peak}} \approx i_1(t_1)$	10.766 mA	10.474 mA	$+2.79\%$

Tables 1 and 2 together with Fig. 1 confirm the three claims of the reduction: the nearby critical instants differ by at most 0.194% of T_{on} , the commutation interval is similarly short, and the small-slope approximation introduces prediction errors below 3% .

6 NMOS Peak-Current Derivation

6.1 Governing Equations and Reduced-Model Assumptions

We set $t_{\text{on}} \equiv t_D$, the instant at which the drain current reaches its maximum. During conduction, KVL in the drain and gate loops gives

$$V_{\text{L1}} = V_{\text{oc}} - R_{\text{in}}(i_1 + i_2) - V_{\text{ds}}, \quad (44)$$

$$V_{\text{L2}} = V_{\text{oc}} - R_{\text{in}}(i_1 + i_2) - V_{\text{gs}}. \quad (45)$$

The MOSFET gate draws only negligible current through its capacitances, so

$$i_2 \approx 0 \quad \Rightarrow \quad i_1 \approx i_{\text{ds}}. \quad (46)$$

With (46), Eqs. (44)–(45) become

$$V_{\text{L1}} = V_{\text{oc}} - R_{\text{in}}i_{\text{ds}} - V_{\text{ds}}, \quad (47)$$

$$V_{\text{L2}} = V_{\text{oc}} - R_{\text{in}}i_{\text{ds}} - V_{\text{gs}}. \quad (48)$$

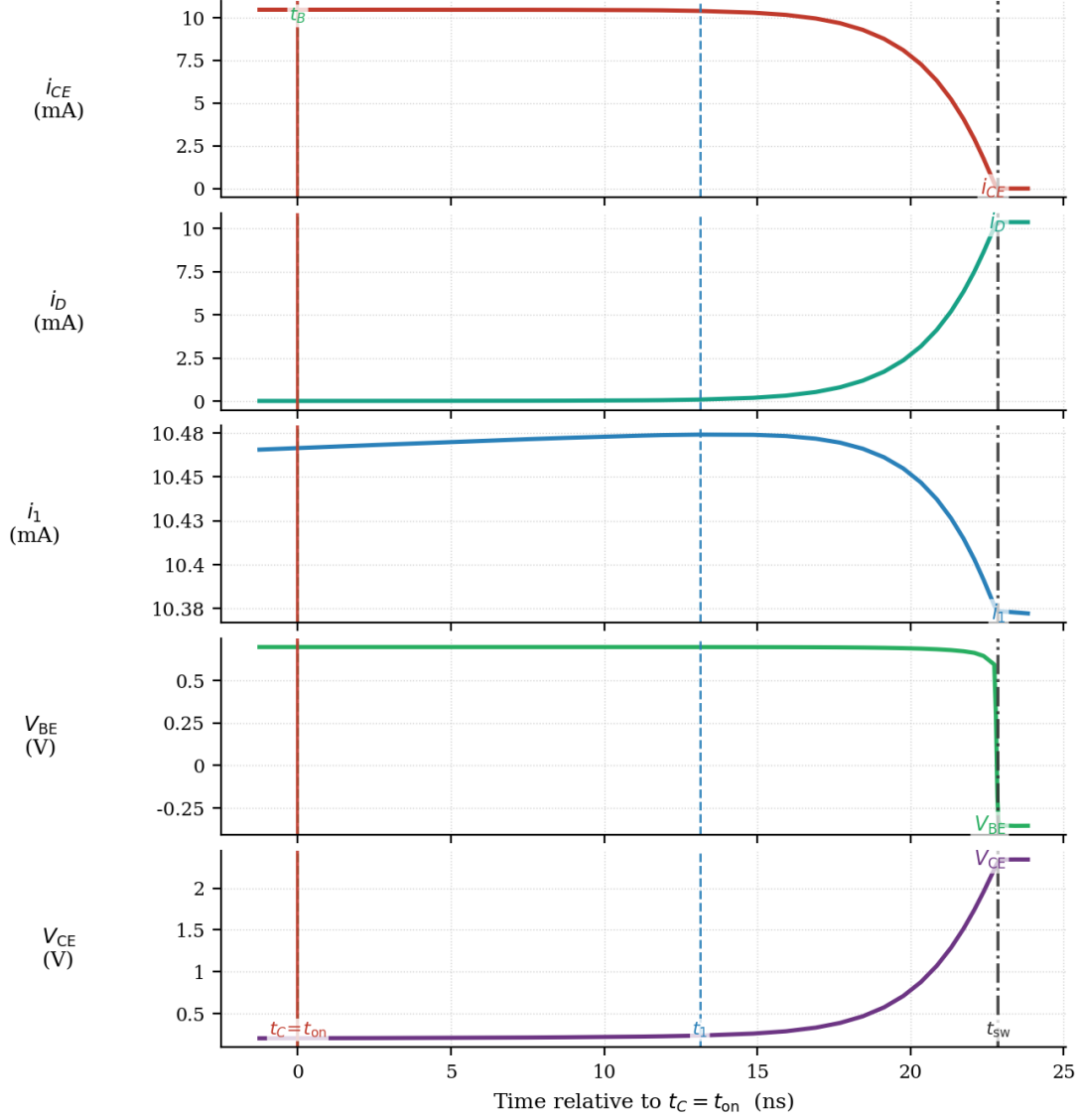


Figure 1: LTspice waveforms near the end of the BJT turn-on interval. Dashed markers identify t_B , $t_C = t_{on}$, and t_1 .

Eliminating V_{L1} between (47) and (48) via $V_{L2} = -aV_{L1}$ from (5) gives the gate-voltage constraint

$$V_{gs} = (1 + a)(V_{oc} - R_{in}i_{ds}) - aV_{ds}. \quad (49)$$

The NMOS operates in the triode region, with drain current

$$i_{ds} = \beta_n V_{ds} \left(V_{gs} - V_{th} - \frac{V_{ds}}{2} \right). \quad (50)$$

Defining the coefficients

$$b_1 \equiv 2a + 1, \quad b_2 \equiv (1 + a)R_{in}, \quad b_3 \equiv (1 + a)V_{oc} - V_{th}, \quad (51)$$

and substituting (49) into (50), solving for i_{ds} , gives the explicit rational function

$$i_{ds}(V_{ds}) = \frac{\beta_n V_{ds} \left[b_3 - \frac{b_1}{2} V_{ds} \right]}{1 + \beta_n b_2 V_{ds}}. \quad (52)$$

6.2 Peak Condition and Closed-Form Solution

Since i_{ds} in (52) depends on time only through $V_{\text{ds}}(t)$, the chain rule gives

$$\frac{di_{\text{ds}}}{dt} = \frac{\partial i_{\text{ds}}}{\partial V_{\text{ds}}} \frac{dV_{\text{ds}}}{dt}. \quad (53)$$

The factor dV_{ds}/dt in (53) is positive at $t = t_{\text{D}}$, by the same argument used for the BJT. At the drain-current peak, di_{ds}/dt passes through zero from positive to negative, so $V_{\text{L1}} = L_1 di_{\text{ds}}/dt$ is positive and decreasing there ($dV_{\text{L1}}/dt < 0$). Differentiating (47) with respect to time and evaluating at t_{D} where $di_{\text{ds}}/dt = 0$ gives

$$\left. \frac{dV_{\text{ds}}}{dt} \right|_{t=t_{\text{D}}} = -R_{\text{in}} \underbrace{\left. \frac{di_{\text{ds}}}{dt} \right|_{t=t_{\text{D}}}}_{=0} - \left. \frac{dV_{\text{L1}}}{dt} \right|_{t=t_{\text{D}}} = - \left. \frac{dV_{\text{L1}}}{dt} \right|_{t=t_{\text{D}}} > 0. \quad (54)$$

The peak condition $di_{\text{ds}}/dt = 0$ is therefore equivalent to

$$\left. \frac{\partial i_{\text{ds}}}{\partial V_{\text{ds}}} \right|_{V_{\text{ds}}=V_{\text{ds}}^*} = 0. \quad (55)$$

Differentiating (52) with respect to V_{ds} gives

$$\frac{\partial i_{\text{ds}}}{\partial V_{\text{ds}}} = \frac{\beta_{\text{n}} \left(b_3 - b_1 V_{\text{ds}} - \frac{\beta_{\text{n}} b_1 b_2}{2} V_{\text{ds}}^2 \right)}{(1 + \beta_{\text{n}} b_2 V_{\text{ds}})^2}. \quad (56)$$

Since $\beta_{\text{n}} > 0$ and the denominator is strictly positive, the peak condition (55) is equivalent to

$$b_3 - b_1 V_{\text{ds}}^* - \frac{\beta_{\text{n}} b_1 b_2}{2} V_{\text{ds}}^{*2} = 0. \quad (57)$$

The physically relevant (positive) root is

$$V_{\text{ds}}^* = \frac{\sqrt{b_1(b_1 + 2\beta_{\text{n}} b_2 b_3)} - b_1}{\beta_{\text{n}} b_1 b_2}. \quad (58)$$

Substituting (58) into (52) gives the peak current

$$i_{\text{peak}} = \frac{\beta_{\text{n}} b_2 b_3 + b_1 - \sqrt{b_1(2\beta_{\text{n}} b_2 b_3 + b_1)}}{\beta_{\text{n}} b_2^2}, \quad (59)$$

which matches the manuscript formula for $b_2 > 0$. The ideal-source case ($R_{\text{in}} = 0$, hence $b_2 = 0$) is a degenerate limit: direct substitution of $b_2 = 0$ into (58) or (59) gives an indeterminate 0/0 form, while the peak-condition equation (57) also reduces from a quadratic equation to a linear one. Therefore, for the ideal-source case, we set $b_2 = 0$ before solving the peak condition. From (52), this gives

$$i_{\text{ds}}(V_{\text{ds}}) = \beta_{\text{n}} V_{\text{ds}} \left[b_3 - \frac{b_1}{2} V_{\text{ds}} \right], \quad (60)$$

for which the peak condition $\partial i_{\text{ds}}/\partial V_{\text{ds}} = 0$ yields directly

$$V_{\text{ds}}^* = \frac{b_3}{b_1} = \frac{(1+a)V_{\text{oc}} - V_{\text{th}}}{2a+1}, \quad (61)$$

and

$$i_{\text{peak}} = \frac{\beta_{\text{n}} b_3^2}{2b_1} = \frac{\beta_{\text{n}} [(1+a)V_{\text{oc}} - V_{\text{th}}]^2}{2(2a+1)}, \quad (62)$$

recovering the manuscript result. The same expression is also obtained by taking the Taylor expansion of (59) at $b_2 = 0$.